# Domain-Specific Image Generation with LoRA

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Part 2 findings - DRAFT: full experiment or submission evidence is still missing


## Abstract

This experiment compares Stable Diffusion 1.5 with two attention-only LoRA adapters, ranks 4 and 8. Both use the same cleaned dataset, split and update budget. The report reads saved measurement files; it does not invent missing scores. The uploaded Part 2 notebook failed during setup and contains no training results.


## I. Data and experimental setup

I use Naruto BLIP Captions, an anime-frame dataset with automatically generated captions. Exact duplicate pixels, corrupt files and empty captions are removed before the seed-266 90/10 split. Images are converted to RGB and resized to 512 x 512. Test images are reserved for evaluation.

Locally verified preparation: {"corrupt": 0, "duplicate": 0, "empty_caption": 0}; 1098 train and 123 test images. These local data checks are separate from the failed uploaded training run.

Two important limits are narrow anime/character coverage and noisy captions. In the prepared data, some captions repeat words or describe clothing vaguely. These issues can weaken prompt following and limit generalization to photographs or other domains.

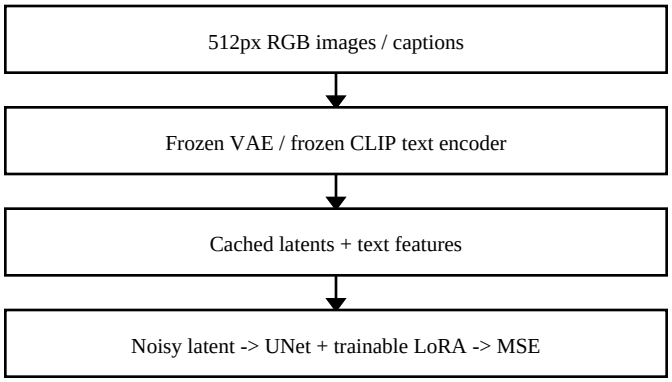


*Fig. 1. Training architecture. Frozen components are run before the UNet to reduce peak memory.*

## II. Controlled training comparison

Both ranks use 1200 AdamW updates at learning rate 0.0001, microbatch 1 and accumulation 4. Only attention LoRA weights update. Base weights stay frozen, gradient checkpointing is enabled, and fp16 is used for the frozen UNet. The lab requires a working single-16-GB-GPU path. A successful 24-GB RTX 4090 setup alone does not demonstrate that memory target.

No completed CUDA training measurements are present. Parameter counts, final loss, peak VRAM and GPU-hours must come from the new run; they cannot be inferred from the installation log.

Training loss curves are not available. The uploaded Part 2 run stopped before its first optimizer update.

## III. Generation and evaluation

Twenty prompts are frozen before training: ten held-out captions, five out-of-domain requests and five compositional requests. Base, rank 4 and rank 8 each generate seeds 0 and 1 using the same sampler, step count and guidance settings: 120 images in total. The safety checker remains enabled. Eight grids compare the models side by side.

Sampling protocol: {"scheduler": "DDIMScheduler", "steps": 20, "guidance_scale": 7.5, "eta": 0.0, "status": "planned; no generated-image results yet"}

KID compares the twenty in-domain generations per model with the test images; lower is better, but the estimate is noisy at this size and may be negative. CLIP score measures image-prompt alignment. LPIPS measures diversity across the two seeds for each prompt. None of these replaces checking whether characters, counts and relationships are actually correct.

Automatic quality metrics are pending. No image-quality conclusion can be made from the failed uploaded Part 2 notebook.

## IV. Human evaluation and model choice

The program creates 25 blinded comparisons with randomized left/right placement. Two team members and at least one person outside the team must independently judge domain style and prompt match. The reported win rate is final-run wins divided by all judgments, with ties reported separately. Human judgments are not generated by an LLM.

Human evaluation is pending, as requested. Do not submit an invented win rate.

Final-run selection is pending inspection of the actual generated images and metric trade-offs.

## V. Failure analysis

At least two confirmed failure modes still need to be recorded after viewing the outputs. Low alignment, ignored object counts, repeated poses and out-of-domain style drift are candidates to inspect, not findings from a run that never trained.

## VI. Safety, licensing and memorization

The Naruto dataset card does not include an explicit license field. Instructor approval of a dataset option does not itself grant copyright or redistribution rights. Anime frames may contain copyrighted characters and designs, and generated images may reproduce similar features. Before uploading adapters or samples, confirm the intended educational sharing is permitted. The base SD 1.5 model has its own CreativeML Open RAIL-M conditions; they do not resolve source-image rights.

The diffusion pipeline keeps its NSFW checker and records its flag for every generation. A filtered output remains counted rather than being silently replaced. The checker can make mistakes, so manually inspect outputs before sharing. Avoid unsafe, deceptive or real-person-targeted prompts. Use private artifact repositories by default and share access with the graders as required. Private sharing reduces exposure but does not create a license.

The memorization check embeds ten seed-0 in-domain generations from each adapter with CLIP, compares them with all training images, and shows the three highest-similarity pairs. High cosine similarity is a warning for inspection, not proof that an image was copied. Low similarity is not proof that memorization is absent. The check is limited by ten samples, a single embedding model and a small training set. A stronger check would use more prompts, perceptual hashes and direct visual review.

Do not publish rater identities or authentication tokens. The human-rating form uses anonymous IDs, and the token is entered using a hidden prompt. Both team members still need to inspect the code, run the experiment and explain the modeling and safety decisions during the demo.

## Submission status

This is a Part 2 section draft, not the required final combined report for all three parts. The final submission also needs team contributions, AI disclosure, discussion recordings, the tag commit links, and the Kaggle link. Pending items are listed below so incomplete work is not mistaken for a finished submission.

Pending: run1 cuda training

Pending: run2 cuda training

Pending: 120 images same protocol

Pending: 8 comparison grids

Pending: kid clip lpips for all models

Pending: final run with explanation

Pending: 2 visually reviewed failure modes

Pending: 25 blind pairs 3 real raters

Pending: memorization 10 images and 3 pairs

Pending: both adapters and image dataset uploaded

Pending: team details recorded

Pending: license and redistribution reviewed

## References

[1] DATA 266 Lab 2 requirements, pp. 8-9 and 11-12, provided by the course.

[2] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022. arxiv.org/abs/2112.10752.

[3] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models, 2021. arxiv.org/abs/2106.09685.

[4] Carlini et al. Extracting Training Data from Diffusion Models, 2023. arxiv.org/abs/2301.13188.